

BRIEF REPORT

Grading differentiation in cutaneous squamous cell carcinoma: A modified Delphi consensus panel

Poor differentiation in cutaneous squamous cell carcinoma (CSCC) is strongly associated with poor outcomes.¹ Despite this, there is no standardized grading system, and it is inconsistently recorded in clinical practice.² Furthermore, CSCC differentiation grading by dermatopathologists and Mohs surgeons is not always consistent.^{3,4} Given these concerns, differentiation was removed from the American Joint Committee on Cancer staging system.² A Research and Development/University of California Los Angeles (RAND/UCLA)-style modified Delphi panel was performed to improve standardization of invasive CSCC differentiation grading.⁵

The panel included 8 dermatopathologists and 8 fellowship-trained Mohs surgeons with > 5 years of experience, national expertise in CSCC (publications and/or national talks), and interpreting > 250 CSCC cases annually. Panelists scored a series of statements from 1 (strongly disagree) to 9 (strongly agree) using a REDCap survey. Median rating and disagreement index (DI) (DI = interpercentile range/interpercentile range adjusted for symmetry; DI ≥ 1 represents lack of consensus) were calculated for each statement. The panelists then met in person to review the statements and initial results. Based on this discussion, statements were revised and the REDCap survey repeated. Statements were considered appropriate with panel consensus when they had a median rating of 7-9 and a DI < 1.

Fourteen panelists attended the meeting and 15 completed both surveys (Supplementary Table I, available via Mendeley at <https://data.mendeley.com/datasets/g5hb4bj9yc/1>). Of the 28 statements included in the 2nd round survey, 27 were considered appropriate with panel consensus. In terms of grading structure (Table I), the panelists agreed that differentiation should be recorded by dermatopathologists, pathologists, and Mohs surgeons on any CSCC specimen with adequate tissue to make this assessment. While differentiation should be included in CSCC staging systems, there should be continued efforts toward developing a standardized, reliable, valid, and user-friendly grading system that correlates with outcomes. The panelists agreed that differentiation should be recorded as well, moderately, or poorly differentiated with intermediate grades avoided when possible. A dichotomous system may be explored in

the future but has not yet been shown to correlate with outcomes and would lose moderate differentiation as an intermediate risk factor. In tumors with biphasic differentiation, the least differentiated appearing portion of the tumor should be considered and graded.

Keratinization (including pearls, horn cysts, and to a lesser extent intracellular keratin, but not parakeratosis or keratin on the tumor surface alone), cellular atypia (including pleomorphism and spindling, but not mitoses), and tumor architecture (including growth pattern, cellular cohesion, and intercellular bridging) should all be considered when grading differentiation (Table II). Additionally, acantholysis and desmoplasia should be recorded. For each of these features, the entire tumor should be assessed and described, paying careful attention to the least differentiated appearing portion. If a CSCC cannot be identified without immunohistochemical staining, it should be considered poorly differentiated.

In conclusion, this modified Delphi study identifies key elements to improving the standardization of CSCC differentiation grading and emphasizes the importance of incorporating differentiation into CSCC prognostication. Clinicians should aim to perform biopsies with adequate substance and depth for grading CSCC differentiation. Further research is needed to develop a reliable, valid, user-friendly grading system that correlates with disease outcomes.

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Table 1. Round 2 survey results regarding structure of cutaneous squamous cell carcinoma differentiation grading

Statement	Median	DI
Differentiation should be determined in all cases of CSCC, as long as there is adequate tissue available to make this assessment. In specimens that do not contain enough tissue to make a determination (such as superficial shave biopsies or Mohs layers with minimal residual tumor), as much information as possible should be recorded about the histologic features relevant to differentiation.	9.0	0.1
Dermatopathologists, pathologists, and Mohs surgeons should all aim to consistently determine CSCC differentiation on specimens that contain adequate tissue to make this assessment. In specimens that do not contain enough tissue to make a determination, these specialists should record as much information as possible about the histologic features relevant to differentiation.	9.0	0.1
CSCC differentiation may be determined on permanent or frozen sections processed vertically or <i>en face</i> . However, it may not be possible to fully evaluate differentiation on each of these specimens depending on the amount and depth of visible tumor. For example, it may be difficult to adequately assess differentiation on a superficial shave biopsy. For each type of specimen, effort should be made to record as much information as possible about the histologic features relevant to differentiation.	8.0	0.1
Differentiation should be included in CSCC staging systems. There should be continued efforts toward developing a standardized, reliable, and valid system of determining differentiation that correlates with disease outcomes.	9.0	0.1
The system for determining CSCC differentiation should be as simple and user friendly as possible to encourage consistent use in clinical practice.	9.0	0.0
CSCC differentiation should be described as well, moderate, or poorly differentiated.	8.0	0.3
Intermediate differentiations (such as poorly to moderately differentiated) should be avoided when possible.	8.0	0.5
A dichotomous system (such as well/moderately differentiated [low grade] and poorly/undifferentiated [high grade]) may be explored in the future if it can be shown to be reliable and valid. This type of system has the potential to reduce ambiguity and increase reliability, but has not yet been shown to correlate with disease outcomes. In addition, moderate differentiation may be an intermediate risk factor for poor outcomes, and this designation would be lost in a dichotomous system.	8.0	0.3
Some tumors may have “biphasic” differentiation. For example, the majority of the tumor may appear to be well differentiated, with a minor focus of poor differentiation. The worst appearing portion of the tumor* should be considered when assessing the overall differentiation. It is also helpful to describe the histologic features of both portions of the tumor and an approximation of the percentage of tumor comprised by each.	8.0	0.3

Median appropriateness rating (1-3 = not appropriate, 4-6 = uncertain, 7-9 = appropriate) and disagreement index (DI = interpercentile range [IPR]/interpercentile range adjusted for symmetry [IPRAS]; DI of greater than or equal to 1 represents lack of consensus among participants).

CSCC, Cutaneous squamous cell carcinoma; DI, disagreement index.

*Least differentiated appearing portion of the tumor.

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Table II. Round 2 survey results regarding histologic features to be included in cutaneous squamous cell carcinoma differentiation grading

Statement	Median	DI
Keratinization should be considered when assessing CSCC differentiation.	9.0	0.1
Keratin pearls and horn cysts should be considered as keratinization. To a lesser degree, intracellular keratin formation (pink, glassy keratin within tumor cells) should be considered.	9.0	0.1
Parakeratosis or keratin on top of the tumor alone should not be considered as keratinization.	8.0	0.1
The entire visible tumor should be used to assess keratinization, and attention should be paid to the worst appearing portion of the tumor* (area with the least degree of keratinization).	9.0	0.1
Some tumors have “biphasic” keratinization. For example, the majority of the tumor may have sparse or absent keratinization, with a minor focus that has prominent keratinization. The worst appearing portion of the tumor* should be considered when assessing keratinization. It is also helpful to describe both portions of the tumor and an approximation of the percentage of tumor comprised by each.	8.0	0.1
Cellular atypia should be considered when assessing CSCC differentiation.	9.0	0.3
Pleomorphism (variability in size and shape of tumor cells and their nuclei) should be considered when assessing cellular atypia.	9.0	0.3
Presence of spindled keratinocytes is an important feature of cellular atypia that should be considered when assessing CSCC differentiation.	9.0	0.1
Number of mitotic figures should not be considered when assessing CSCC differentiation, as it is not a reliable measure of cellular atypia in CSCC and is difficult and time consuming to measure in routine clinical practice.	8.0	0.1
The entire visible tumor should be used to assess cellular atypia, and careful attention should be paid to the worst appearing portion of the tumor*.	9.0	0.1
Some tumors may have “biphasic” cellular atypia. For example, the majority of the tumor may have minimal cellular atypia, with a minor focus that has prominent cellular atypia. The worst appearing portion of the tumor* should be considered when assessing cellular atypia. It is also helpful to describe both portions of the tumor and an approximation of the percentage of tumor comprised by each.	8.0	0.3
Tumor architecture should be considered when assessing CSCC differentiation.	8.0	0.3
The entire visible tumor should be used to assess tumor architecture, and attention should be paid to the worst appearing portion of the tumor*.	9.0	0.1
Tumor growth pattern (whether borders are pushing or infiltrative with single cell spread and/or tumor budding) is an important aspect of tumor architecture that should be considered when assessing CSCC differentiation.	7.0	0.7
Cellular cohesion and intercellular bridging (presence of desmosomal bridges connecting tumor cells) are important aspects of tumor architecture that should be considered when assessing CSCC differentiation.	7.0	0.7
Acantholysis (which is a measure of cellular cohesion) should be recorded when present.	7.0	0.3
Whether tumor islands display maturation is an important aspect of tumor architecture that should be considered when assessing CSCC differentiation.	6.0	0.7
Desmoplasia (formation of a fibrotic stroma around the tumor cells) should be considered in assessing CSCC differentiation.	7.0	0.7
If a tumor cannot be identified as CSCC without immunohistochemical staining, it should be considered poorly differentiated.	9.0	0.0

Median appropriateness rating (1-3 = not appropriate, 4-6 = uncertain, 7-9 = appropriate) and disagreement index (DI = interpercentile range [IPR]/interpercentile range adjusted for symmetry [IPRAS]); DI of greater than or equal to 1 represents lack of consensus).

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*Least differentiated appearing portion of the tumor.

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Funding sources: This study was funded by a research grant from the Dermatology Foundation.

Patient consent: N/a.

IRB approval status: Reviewed and approved by the Ohio State University Institutional Review Board; approval #2023C0129.

Key words: cellular atypia; cutaneous squamous cell carcinoma; differentiation; histologic grade; keratinization; modified Delphi panel; tumor architecture.

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Conflicts of interest

Sean R. Christensen has served as a consultant for Castle Biosciences; Emily S. Ruiz serves as a consultant for Regeneron Inc., Checkpoint Therapeutic, Feldan Pharmaceuticals, and Merck Pharmaceuticals. She serves

as a Principal Investigator/Co-Investigator for the following companies: Regeneron (PI/Co-I), Merck (Co-I), and Castle Biosciences (PI/Co-I). She is on the Board of Directors of Skin Cancer Outcomes Consortium (SCOUT); Allison Vidimos has been an investigator for Castle and on the Advisory Committee for Inhibitor Therapeutics; David R. Carr has been an investigator for Regeneron. The other authors have no conflicts of interest to declare.

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<https://doi.org/10.1016/j.jaad.2025.05.003>